

Objective: I can solve equations where the inverse power results in a quadratic function, handle extraneous solutions, and manipulate literal equations.

Part 1: When the Result is Quadratic

Sometimes, eliminating the radical leaves you with an x^2 term. You must set the equation to zero and solve.

Example A: Teacher Model

1. Factoring Required

$$\sqrt[3]{x^2 - 4x} = \sqrt[3]{12}$$

2. Quadratic Formula Required

$$\sqrt[3]{x^2 - 6x} = \sqrt[3]{9}$$

Practice: Resulting in Quadratics

1. Variable on Outside

$$\sqrt{2x + 15} = x$$

2. Cube Root Quadratic

$$2(x^2 + 5x)^{\frac{1}{3}} = 4$$

Part 2: The Even Numerator "Two Case" Scenario

When the numerator of the exponent is **EVEN** (e.g., $\frac{2}{3}$), taking the reciprocal root creates a $\pm$ situation.

Example B

1. Standard Even Root

$$(x - 1)^{\frac{2}{3}} = 9$$

2. Complex Even Root

$$2(x^2 - 14)^{\frac{2}{3}} = 50$$

Part 3: Literal Equations

Isolate the specific variable requested. Do not plug in numbers.

Practice: Isolating Variables

5. Biology (Kleiber's Law)

Solve this equation for **m**.

$$M = k \cdot m^{\frac{3}{4}}$$

Part 4: Challenge Application

Planetary Motion: Kepler's Third Law states that the square of the orbital period P (in years) is equal to the cube of the average distance a (in AU) from the Sun:

$$P = a^{\frac{3}{2}}$$

Problem: A newly discovered asteroid has an orbital period P of 125 years.

1. Write an equation and solve for the average distance a .
2. If a planet is 4 times further from the sun than the asteroid ($4a$), what is its period?